FYP

Chapter: 10 – Formal Reports

Formal Reports

In comparison to informal documents, formal documents usually:

(1) cover more complicated projects

(2) are longer than their informal counterparts

(3) have a more diverse set of readers

Headings in a Report

FIRST-LEVEL HEADING
(all caps, bold, on a line by itself)

Second-level heading
(initial cap only, bold, on a line by itself)

Third-Level heading (initial cap only, bold, followed by two spaces, as part of the first line of the paragraph)

Second-level Heading in a Report

Third-level heading in a short report

Decimal Headings

1 xxxxxxxxxxxxxx

1.1 xxxxxxxxxxx

1.1.1 xxxxxxxxxxx

1.1.2 xxxxxxxxxxx

1.2 xxxxxxxxxxx

1.2.1 xxxxxxxxxxx

1.2.2 xxxxxxxxxxx

2 xxxxxxxxxxxxxx

2.1 xxxxxxxxxxx

2.1.1 xxxxxxxxxxx

2.1.2 xxxxxxxxxxx

2.2 xxxxxxxxxxx

3 xxxxxxxxxxxxxx

Parts of a Formal Reports and FYP

1. Cover/title page
2. Letter or memo of transmittal
3. Table of contents
4. List of illustrations
5. Executive summary
6. Introduction
7. Discussion sections
8. Conclusions and recommendations
9. End material

Formal Report	FYP
Cover/title page Letter or memo of transmittal Table of contents List of illustrations Executive summary Introduction Discussion sections Conclusions and recommendations End material	Title Page Undertaking Table of contents List of Tables List of Figures Abstract Introduction (Chapter-1) Literature Review (Chapter-2) Requirements and Design (Chapter: 3) Implementation and Test Scores (Chapter: 4) Results and Analysis (Chapter: -5) Conclusions (Chapter-6) References Appendix

Cover/Title Page

- Project title
- Your client's or recipient's name ("Prepared for . . .")
- Your name and/or the name of your organization ("Prepared by . . .")
- Date of submission

Use a visual only if it reinforces a main point.

Letter/memo of Transmittal

- Place Letter/memo immediately after the title page
- Include a major point from the report (Why you are writing/ What exactly of importance is within it
- Follow letter and memo conventions (e.g. single-spacing; use only one page)

MEMO

TO: Karrie Camp, Vice President for Human Resources

FROM : Abe Andrews, Personnel Assistant *aa*

SUBJECT: Report on Flextime Pilot Program at Boston Office

DATE: March 18, 2012

As you requested, I have examined the results of the six-month pilot program to introduce flextime to the Boston office. This report presents my data and conclusions about the use of flexible work schedules.

To determine the results of the pilot program, I asked all employees to complete a written survey. Then I followed up by interviewing every fifth person on an alphabetical list of office personnel. Overall, it appears that flextime has met with clear approval by employees at all levels. Productivity has increased and morale has soared. This report uses the survey and interview data to suggest why these results have occurred and where we might go from here.

I enjoyed working on this personnel study because of its potential impact on the way M-Global conducts business. Please give me a call if you would like additional details about the study.

M-GLOBAL, INC.

M-Global Inc | 127 Rainbow Lane | Baltimore MD 21202 | 410.555.8175

Table of Contents

- It acts as an **outline**
- It should be a complete and accurate **listing** of the main and minor topics covered in the report.
- You don’t want just a brief and sketchy outline of major headings.
- An effective table of contents fleshes out the details, so your readers know exactly what is covered in each section – saves their time and helps them find the information they want and need.

TABLE OF CONTENTS

List of Illustrations

- If your report contains several tables and figures, you will need to provide a list of illustrations.
- This list can be included below your table of contents, if there is room on the page, or on a separate page.
- As with the table of contents, your list of illustrations must be clear and informative.

Abstract (or Executive Summary)

- The abstract is a brief overview of the report's key points geared towards a varied audience from a low-tech reader to managers, supervisors, and highly placed executives.

- They need your help in two ways:
  - They need information quickly
  - They need it presented in a low-tech terminology
- You can achieve both these objectives through an abstract or executive summary.

Introduction

- Not to summarize the report
- Give information on, the report's:
  1. **Purpose:** State your purpose, as the first part of the introduction
  2. **Scope:** detail and description of your project should be written clearly
  3. **Format:** a brief preview of the main sections that follow

Conclusion/Recommendation

- Providing your readers with a sense of closure
- Discuss results based on findings of your study.
- Recommendations are actions suggested on the basis of your conclusion.
- Executive summary consists of a brief description of the most important conclusion and recommendation, whereas the conclusion/recommendation section is an expanded version of the executive summary.

Appendices

- A final optional component is an appendix.
- It allows you to include any additional information (surveys, results, tables, figures, previous report findings, relevant letters or memos, etc.) that you have not built in your report's main text.
- The contents of your appendix should not be of primary importance, which needs to be the part of the body of the report.
- An appendix is a perfect place to file nonessential data that provides documentation for future reference.

References (or Bibliography)

- The two parts to referencing are:
  - **citations** in the text of the report
  - **a list of references** in the final section
- A citation shows that information comes from another source. The reference list gives the details of these sources. You need to use in-text citations and provide details in the references section when:
  - you incorporate information from other sources; e.g.:
  - factual material
  - graphs and tables of data
  - pictures and diagrams

1. Writing an ABSTRACT

FORMAL REPORT WRITING

“ Qualities of a Good Abstract

- Uses one well developed paragraph which is unified, coherent, concise, and able to stand alone.
- Uses an introduction/body/conclusion structure which presents the article, paper, or report's purpose, method, Results, conclusion(s), and recommendations in that order.
- Follows strictly the chronology of the article, paper, or report.
- Provides logical connections (or transitions) between the information included.
- Adds no new information, but simply summarizes the report.
- Is understandable to a wide audience.
- Any major restrictions or limitations on the results should be stated, if only by using "weasel-words" such as "might", "could", "may", and "seem".

Types of Abstracts

Descriptive: (less than 100 words)

- Indicates the type of information found in the work.
- Makes no judgments about the work, nor does it provide results or conclusions of the research.

Example (Descriptive): The two most common abstract types—descriptive and informative—are described and examples of each are provided.

Informative: (more than 250 words)

- The writer presents and explains all the main arguments and the important results and evidence present in the complete report/article/paper/book.
- Do not critique or evaluate a work, they do more than describe it.

Example (Informative): Abstracts present the essential elements of a longer work in a short and powerful statement. The purpose of an abstract is to provide prospective readers the opportunity to judge the relevance of the longer work to their projects. Abstracts also include the key terms found in the longer work and the purpose and methods of the research. Authors abstract various longer works, including book proposals, dissertations, and online journal articles. There are two main types of abstracts: descriptive and informative. A descriptive abstract briefly describes the longer work, while an informative abstract presents all the main arguments and important results. This handout provides examples of various types of abstracts and instructions on how to construct one.

Executive Summaries vs. Abstracts

- Executive summaries go by so many different names. Sometimes the executive summary is called an Abstract. You usually find that designation in scientific papers and academic efforts. You can also call the Executive Summary simply a Summary.

- Abstracts differ from executive summaries, because abstracts are usually written for a scientific or academic purpose.

Abstract Components (HOW)

Brief Background	(optional)
Reason for writing	What is the importance of the research? Why would a reader be interested in the larger work?
Problem	(Optional) What problem does this work attempt to solve? What is the scope of the project? What is the main argument/thesis/claim?
Methodology	An abstract of a scientific work may include specific models or approaches used in the larger study.
Reults/ Findings/ Implementation	An abstract of a scientific work may include specific data that indicates the results of the project.
Conclusion and Implications	What changes should be implemented as a result of the findings of the work?

Sample Abstract

This study's **objective** was to/focuses on determine the strangeness measurements for red, green, and blue quarks. The Britt-Cushman **method** for quark analysis exploded/explode a quarkstream in a He gas cloud. **Results** indicate that both red and green quarks had a strangeness that differed by less than 0.453 x 10-17 Zabes/m2 for all measurements. Blue quarks remained immeasurable, since their particle traces bent into 7-tuple space. This study's **conclusions** indicate that red and green quarks can be used interchangeably in all He stream applications, and **further studies** must be done to measure the strangeness of blue quarks.

Compound Sentence Segmentation and Sentence Boundary Detection in Urdu

The raw Urdu corpus comprises of irregular and large sentences which need to be properly segmented in order to make them useful in Natural Language Engineering (NLE). This makes the Compound Sentences Segmentation (CSS) timely and vital research topic. The existing online text processing tools are developed mostly for computationally developed languages such as English, Japanese and Spanish etc., where sentence segmentation is mostly done on the basis of delimiters. Our proposed approach uses special characters as sentence delimiters and computationally extracted sentence-endletters and sentence-end-words as identifiers for segmentation of large and compound sentences. The raw and unannotated input text is passed through preprocessing and word segmentation. Urdu word segmentation itself is a complex task including knotty problems such as space insertion and space deletion etc. Main and subordinate clauses are identified and marked for subsequent processing. The resultant text is further processed in order to identify, extract and then segment large as well as compound sentences into regular Urdu sentences. Urdu computational research is in its infancy. Our work is pioneering in Urdu CSS and results achieved by our proposed approach are promising. For experimentation, we used a general genre raw Urdu corpus containing 2616 sentences and 291503 words. We achieved 34% improvement in reduction of average sentence length from 111 w/s to 38 w/s (words per sentence). This increased the number of sentences by almost three times to 7536 shorter and computationally easy to manage sentences. Resultant text reliability and coherence are verified by Urdu language experts.

Urdu Language Translator using Deep Neural Network

It was clearly seen that the proposed model shows the high accuracy when the input is recorded audio and it shows poor performance with real time input. While one HTTP request per input transcription produced English translation for Text to Text translation using Python Text Blob library. This paper proposes an interactive Urdu to English language speech translator using deep Neural Network. ASR module in proposed pipeline is composed of deep neural network and is simpler as compared to traditional ASR which requires complex hand engineering like feature extraction and resources like phoneme dictionary. The proposed speech recognition model out performs traditional automatic speech recognition systems in efficiency, simplicity and robustness. The final output was achieved with a delay of no more than 30 seconds. Furthermore, we have tested and provided some statistical findings, the result shows that value updating for neural network layer's bias, standard deviation when Adam optimizer parameters are set as follows: beta1=0.9, beta2=0.9 and learning rate =0.01 meanwhile dropout rate was kept to 5% to offer regularization and observed value for scalar maximum lies between 0 and 0.08. There is a little deviation at 0.05 step, value decreases and afterwards that bias maximum scalar increases with positive values and finally increases exponentially at later stages of training further results are discussed in experiment section respectively.

Urdu Language Translator using Deep Neural Network

Purpose: This paper proposes an interactive Urdu to English language speech translator using deep Neural Network. ASR module in proposed pipeline is composed of deep neural network and is simpler as compared to traditional ASR which requires complex hand engineering like feature extraction and resources like phoneme dictionary.

Method: It was clearly seen that the proposed model shows the high accuracy when the input is recorded audio and it shows poor performance with real time input. While one HTTP request per input transcription produced English translation for Text to Text translation using Python Text Blob library.

Result: The final output was achieved with a delay of no more than 30 seconds. Furthermore, we have tested and provided some statistical findings, the result shows that value updating for neural network layer's bias, standard deviation when Adam optimizer parameters are set as follows: beta1=0.9, beta2=0.9 and learning rate =0.01 meanwhile dropout rate was kept to 5% to offer regularization and observed value for scalar maximum lies between 0 and 0.08. There is a little deviation at 0.05 step, value decreases and afterwards that bias maximum scalar increases with positive values and finally increases exponentially at later stages of training further results are discussed in experiment section respectively.

Conclusion: The proposed speech recognition model out performs traditional automatic speech recognition systems in efficiency, simplicity and robustness.